Italic lanes for the base-anchor candidate X

A *lane* is a strip at the italic angle θ (12° italic, 11.5° semibold italic) with a left and a right edge. A line at the italic angle through a point (x, y) meets the baseline at $x - y\tan\theta$, so a lane is given by where its edges meet the baseline, and its midline at height Y is the midpoint of those two values plus $Y\tan\theta$. For a glyph with outline $\mathcal{O}$ and advance width W:

lane	left edge	right edge	midline at Y	
advance lane	0	W	$W/2 + Y\tan\theta$	the model
ink lane	$L = \min_{\mathcal{O}}(x - y\tan\theta)$	$R = \max_{\mathcal{O}}(x - y\tan\theta)$	$C + Y\tan\theta,\ C = (L+R)/2$	the mark-anchor model's axis
ideal lane	L^*	R^*	$(L^* + R^*)/2 + Y\tan\theta$	where the glyph lives; not computed

The ink lane is the narrowest strip at the italic angle that contains the outline: its edges are lines at the italic angle that just touch the outline, as X_{min} and X_{max} are upright lines that just touch it. With $\theta = 0$, $L = X_{\text{min}}$ and $R = X_{\text{max}}$, so in roman both lanes give the bbox midpoint, the roman candidate; only the italic styles are compared. Italic n: $W = 518$, $L = 17$ (the entry serif at the top left), $R = 502$ (the outstroke at the bottom right), so $C = 259 = W/2$ and the lanes agree. They differ where the ink sits off-centre in the advance width.

The candidate is $X =$ midline at the candidate's Y, minus K (semibold italic: $K = 150$ above, -12 below; italic: 0). Y is the same for both lanes.

Summary

δX = anchor X minus candidate X. *designer*: the anchors of the original fonts (the fontmetrics); they were probably placed by advance width, so they favour the advance lane. *no derived or precomposed*: without the letters of `data/derived_letters.py` (which inherit from their base letter) and canonically decomposed letters. *curated*: the human-curated base anchors (`data/fontanchors_human/`, `bases` by codepoint), with the candidate from the original glyph. Columns per lane: median and mean $|\delta X|$, how many $|\delta X| \le 10$, median signed δX. *closer*: how many anchors each lane's candidate is closer to (rounded).

style	anchors	class	n	advance lane				ink lane				closer		
				med	mean	$\le$10	sgn	med	mean	$\le$10	sgn	adv	ink	tie
it	designer, all	a	82	26.4	69.4	21	-26	27.2	69.9	22	-25	41	37	4
it	designer, all	b	108	18.6	31.4	30	14	22.5	34.6	23	13	62	42	4
it	designer, no derived or precomposed	a	73	21.6	56.5	21	-22	23.7	56.0	22	-19	36	33	4
it	designer, no derived or precomposed	b	91	17.4	30.6	26	12	21.8	32.4	21	11	49	38	4
it	curated	a	30	20.0	35.9	5	1	22.5	40.0	8	-3	16	13	1
it	curated	b	22	24.4	37.3	5	-8	24.5	41.3	5	-10	14	7	1
si	designer, all	a	91	21.9	34.0	17	-10	24.9	36.1	19	-8	54	36	1
si	designer, all	b	109	18.1	25.8	29	-3	22.6	29.4	26	-5	71	35	3
si	designer, no derived or precomposed	a	79	21.6	33.0	14	-7	24.8	33.9	16	-8	45	33	1
si	designer, no derived or precomposed	b	92	17.9	25.5	24	-4	22.0	27.6	21	-5	59	30	3
si	curated	a	22	16.7	34.8	7	1	18.6	40.4	4	-2	13	9	0
si	curated	b	19	27.8	37.3	5	-15	22.6	37.8	4	-20	9	10	0

Table rows

hex, cat: codepoint and Unicode category. it, si: the base alone, with dot above, and with macron below (via CGJ), original font; patch: the same in the patched font. a, b: the designer's above and below anchor X, Y. a? adv, b? adv: candidate from the advance lane; aδ adv, bδ adv: anchor minus that candidate. a? ink, b? ink and aδ ink, bδ ink: the same for the ink lane. Of the two δ cells, the one with the smaller $|\delta X|$ is bold.

Table 1: Lane comparison for A–Z 1/2

hex	0041	0042	0043	0044	0045	0046	0047	0048	0049	004A	004B	004C	004D
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
it	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞF*	*GĠG*	*HḢḤ*	*IİỊ*	*J̇J̈J*	*KK̇K*	*LĿḶ*	*MṀṂ*
patch	*AȦẠ*	*BḂḄ*	*CĊÇ*	*DḊḌ*	*EĖẸ*	*FḞF*	*GĠG*	*HḢḤ*	*IİỊ*	*J̇J̈J*	*KK̇K*	*LĿḶ*	*MṀṂ*
a	505, 850	447, 850	500, 850	396, 850	413, 850	406, 850	513, 850	481, 850	316, 850	392, 850	466, 850	389, 850	557, 850
a? adv	514, 850	459, 850	489, 850	514, 850	444, 850	409, 850	513, 850	517, 850	321, 850	388, 850	499, 850	440, 850	583, 850
aδ adv	-9, 0	-12, 0	**11, 0**	-118, 0	-31, 0	**-3, 0**	**0, 0**	-36, 0	-5, 0	**4, 0**	-33, 0	-51, 0	-26, 0
a? ink	514, 850	441, 850	486, 850	494, 850	429, 850	400, 850	514, 850	510, 850	315, 850	374, 850	485, 850	422, 850	582, 850
aδ ink	-9, 0	**6, 0**	14, 0	**-98, 0**	**-16, 0**	6, 0	-1, 0	**-29, 0**	**1, 0**	18, 0	**-19, 0**	**-33, 0**	**-25, 0**
b	308, -108	238, -108	333, -110	279, -107	269, -107	131, -111	345, -110	320, -107	127, -109	109, -319	292, -110	252, -110	363, -110
b? adv	310, -110	255, -110	285, -110	310, -110	240, -110	205, -110	309, -110	313, -110	117, -110	146, -285	295, -110	236, -110	379, -110
bδ adv	-2, 2	-17, 2	**48, 0**	-31, 3	**29, 3**	-74, -1	36, 0	**7, 3**	**10, 1**	-37, -34	**-3, 0**	**16, 0**	-16, 0
b? ink	310, -110	237, -110	282, -110	290, -110	225, -110	196, -110	310, -110	306, -110	111, -110	132, -285	281, -110	218, -110	378, -110
bδ ink	-2, 2	**1, 2**	51, 0	**-11, 3**	44, 3	**-65, -1**	**35, 0**	14, 3	16, 1	**-23, -34**	11, 0	34, 0	**-15, 0**
si	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞF***	***GĠG***	***HḢḤ***	***IİỊ***	***J̇J̈J***	***KK̇K***	***LĿḶ***	***MṀṂ***
patch	***AȦẠ***	***BḂḄ***	***CĊÇ***	***DḊḌ***	***EĖẸ***	***FḞF***	***GĠG***	***HḢḤ***	***IİỊ***	***J̇J̈J***	***KK̇K***	***LĿḶ***	***MṀṂ***
a	354, 850	267, 850	340, 850	306, 850	273, 850	276, 850	365, 850	361, 850	150, 850	432, 850	336, 850	149, 850	431, 850
a? adv	382, 850	319, 850	341, 850	367, 850	292, 850	269, 850	371, 850	396, 850	194, 850	249, 850	369, 850	310, 850	441, 850
aδ adv	**-28, 0**	-52, 0	**-1, 0**	-61, 0	-19, 0	**7, 0**	-6, 0	**-35, 0**	**-44, 0**	**183, 0**	**-33, 0**	**-161, 0**	-10, 0
a? ink	384, 850	290, 850	336, 850	347, 850	283, 850	265, 850	368, 850	403, 850	202, 850	239, 850	380, 850	314, 850	439, 850
aδ ink	-30, 0	**-23, 0**	4, 0	**-41, 0**	**-10, 0**	11, 0	**-3, 0**	-42, 0	-52, 0	193, 0	-44, 0	-165, 0	**-8, 0**
b	308, -110	268, -110	363, -110	279, -110	279, -110	201, -110	367, -110	365, -110	147, -110	109, -319	342, -110	260, -110	393, -110
b? adv	349, -110	286, -110	308, -110	334, -110	259, -110	236, -110	338, -110	363, -110	161, -110	187, -249	336, -110	277, -110	408, -110
bδ adv	**-41, 0**	-18, 0	**55, 0**	-55, 0	**20, 0**	-35, 0	**29, 0**	**2, 0**	**-14, 0**	-78, -70	6, 0	**-17, 0**	-15, 0
b? ink	351, -110	257, -110	303, -110	314, -110	250, -110	232, -110	335, -110	370, -110	169, -110	177, -249	347, -110	281, -110	406, -110
bδ ink	-43, 0	**11, 0**	60, 0	**-35, 0**	29, 0	**-31, 0**	32, 0	-5, 0	-22, 0	**-68, -70**	**-5, 0**	-21, 0	**-13, 0**

Table 2: Lane comparison for A–Z 2/2

hex	004E	004F	0050	0051	0052	0053	0054	0055	0056	0057	0058	0059	005A
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu
it	*N̈N̤*	*ÖO̤*	*P̈P̤*	*Q̈Q̤*	*R̈R̤*	*S̈S̤*	*T̈T̤*	*Ü Ṳ*	*V̈V̤*	*ẄW̤*	*Ẍ X̤*	*Ÿ Ỵ*	*Z̈Z̤*
patch	*N̈N̤*	*ÖO̤*	*P̈P̤*	*Q̈Q̤*	*R̈R̤*	*S̈S̤*	*T̈T̤*	*Ü Ṳ*	*V̈V̤*	*ẄW̤*	*Ẍ X̤*	*Ÿ Ỵ*	*Z̈Z̤*
a	476, 850	515, 850	426, 850	498, 850	419, 850	402, 850	457, 850	486, 850	481, 850	610, 850	482, 850	435, 850	450, 850
a? adv	514, 850	515, 850	430, 850	515, 850	458, 850	408, 850	453, 850	498, 850	479, 850	635, 850	495, 850	457, 850	470, 850
aδ adv	-38, 0	**0, 0**	**-4, 0**	**-17, 0**	**-39, 0**	-6, 0	4, 0	-12, 0	**2, 0**	-25, 0	-13, 0	-22, 0	**-20, 0**
a? ink	510, 850	510, 850	417, 850	524, 850	465, 850	403, 850	453, 850	486, 850	464, 850	634, 850	495, 850	438, 850	475, 850
aδ ink	**-34, 0**	5, 0	9, 0	-26, 0	-46, 0	**-1, 0**	4, 0	**0, 0**	17, 0	**-24, 0**	-13, 0	**-3, 0**	-25, 0
b	328, -109	310, -113	140, -110		248, -111	204, -111	277, -109	308, -109	263, -110	439, -107	287, -107	270, -109	267, -107
b? adv	310, -110	311, -110	226, -110	272, -294	254, -110	204, -110	249, -110	294, -110	275, -110	431, -110	291, -110	253, -110	266, -110
bδ adv	**18, 1**	**-1, -3**	-86, 0		**-6, -1**	**0, -1**	28, 1	**14, 1**	-12, 0	**8, 3**	-4, 3	**17, 1**	**1, 3**
b? ink	306, -110	306, -110	213, -110	281, -294	261, -110	199, -110	249, -110	282, -110	260, -110	430, -110	291, -110	234, -110	271, -110
bδ ink	22, 1	4, -3	**-73, 0**		-13, -1	5, -1	28, 1	26, 1	**3, 0**	9, 3	-4, 3	36, 1	-4, 3
si	***N̈N̤***	***ÖO̤***	***P̈P̤***	***Q̈Q̤***	***R̈R̤***	***S̈S̤***	***T̈T̤***	***Ü Ṳ***	***V̈V̤***	***ẄW̤***	***Ẍ X̤***	***Ÿ Ỵ***	***Z̈Z̤***
patch	***N̈N̤***	***ÖO̤***	***P̈P̤***	***Q̈Q̤***	***R̈R̤***	***S̈S̤***	***T̈T̤***	***Ü Ṳ***	***V̈V̤***	***ẄW̤***	***Ẍ X̤***	***Ÿ Ỵ***	***Z̈Z̤***
a	346, 850	335, 850	316, 850	348, 850	269, 850	232, 850	307, 850	346, 850	336, 850	450, 850	332, 850	295, 850	320, 850
a? adv	394, 850	390, 850	284, 850	390, 850	315, 850	250, 850	309, 850	364, 850	328, 850	501, 850	362, 850	319, 850	322, 850
aδ adv	-48, 0	**-55, 0**	**32, 0**	**-42, 0**	**-46, 0**	-18, 0	**-2, 0**	**-18, 0**	**8, 0**	-51, 0	**-30, 0**	-24, 0	**-2, 0**
a? ink	380, 850	393, 850	271, 850	393, 850	322, 850	245, 850	296, 850	365, 850	312, 850	475, 850	363, 850	302, 850	317, 850
aδ ink	**-34, 0**	-58, 0	45, 0	-45, 0	-53, 0	**-13, 0**	11, 0	-19, 0	24, 0	**-25, 0**	-31, 0	**-7, 0**	3, 0
b	348, -110	350, -110	240, -110		258, -110	234, -110	300, -110	328, -110	318, -110	479, -110	307, -110	295, -110	287, -110
b? adv	361, -110	357, -110	251, -110	314, -319	282, -110	217, -110	276, -110	331, -110	295, -110	468, -110	329, -110	286, -110	289, -110
bδ adv	-13, 0	**-7, 0**	-11, 0		**-24, 0**	**17, 0**	**24, 0**	**-3, 0**	**23, 0**	**11, 0**	**-22, 0**	**9, 0**	**-2, 0**
b? ink	347, -110	360, -110	238, -110	317, -319	289, -110	212, -110	263, -110	332, -110	279, -110	442, -110	330, -110	269, -110	284, -110
bδ ink	**1, 0**	-10, 0	**2, 0**		-31, 0	22, 0	37, 0	-4, 0	39, 0	37, 0	-23, 0	26, 0	3, 0

Table 3: Lane comparison for a–z 1/2

hex	0061	0062	0063	0064	0065	0066	0067	0068	0069	006A	006B	006C	006D
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
it	*aȧạ*	*bḃḅ*	*cċc̣*	*dḋḍ*	*eėẹ*	*fḟf̣*	*gġg̣*	*hḣḥ*	*iiį*	*jj̇ȷ̣*	*kk̇ḳ*	*ll̇ḷ*	*mṁṃ*
patch	*aȧạ*	*bḃḅ*	*cċc̣*	*dḋḍ*	*eėẹ*	*fḟf̣*	*gġg̣*	*hḣḥ*	*iiį*	*jj̇ȷ̣*	*kk̇ḳ*	*ll̇ḷ*	*mṁṃ*
a	341, 647		343, 645	218, 645	332, 645		360, 645	409, 645			405, 646	292, 890	522, 645
a? adv	380, 645	428, 890	332, 645	434, 890	338, 645	346, 890	376, 645	449, 890	310, 810	302, 810	432, 890	322, 890	529, 645
aδ adv	**-39, 2**		11, 0	**-216, -245**	**-6, 0**		**-16, 0**	**-40, -245**			-27, -244	**-30, 0**	**-7, 0**
a? ink	396, 645	411, 890	344, 645	451, 890	345, 645	365, 890	380, 645	455, 890	321, 810	240, 810	430, 890	327, 890	539, 645
aδ ink	-55, 2		**-1, 0**	-233, -245	-13, 0		-20, 0	-46, -245			**-25, -244**	-35, 0	-17, 0
b	174, -108	214, -108	163, -113	221, -107	158, -108	222, -111	197, -319	229, -111	136, -107	0, -319	223, -107	115, -107	382, -105
b? adv	220, -110	216, -110	171, -110	221, -110	177, -110	89, -319	171, -319	236, -110	115, -110	70, -281	220, -110	110, -110	368, -110
bδ adv	**-46, 2**	**-2, 2**	**-8, -3**	**0, 3**	**-19, 2**	133, 208	26, 0	**-7, -1**	21, 3	-70, -38	**3, 3**	5, 3	14, 5
b? ink	236, -110	199, -110	184, -110	239, -110	185, -110	108, -319	175, -319	243, -110	126, -110	8, -281	218, -110	115, -110	379, -110
bδ ink	-62, 2	15, 2	-21, -3	-18, 3	-27, 2	**114, 208**	**22, 0**	-14, -1	**10, 3**	**-8, -38**	5, 3	**0, 3**	**3, 5**
si	***aȧạ***	***bḃḅ***	***cċc̣***	***dḋḍ***	***eėẹ***	***fḟf̣***	***gġg̣***	***hḣḥ***	***iiį***	***jj̇ȷ̣***	***kk̇ḳ***	***ll̇ḷ***	***mṁṃ***
patch	***aȧạ***	***bḃḅ***	***cċc̣***	***dḋḍ***	***eėẹ***	***fḟf̣***	***gġg̣***	***hḣḥ***	***iiį***	***jj̇ȷ̣***	***kk̇ḳ***	***ll̇ḷ***	***mṁṃ***
a	379, 645		242, 645	218, 645	243, 645		291, 645	469, 645	137, 795		306, 784	174, 890	392, 645
a? adv	251, 645	290, 890	203, 645	309, 890	218, 645	216, 890	220, 645	331, 890	180, 823	192, 823	316, 890	194, 890	407, 645
aδ adv	128, 0		39, 0	**-91, -245**	**25, 0**		71, 0	138, -245	**-43, -28**		-10, -106	-20, 0	**-15, 0**
a? ink	274, 645	281, 890	206, 645	317, 890	217, 645	206, 890	233, 645	331, 890	191, 823	113, 823	314, 890	193, 890	412, 645
aδ ink	**105, 0**		**36, 0**	-99, -245	26, 0		**58, 0**	138, -245	-54, -28		**-8, -106**	**-19, 0**	-20, 0
b	196, -110	234, -110	236, -110	221, -110	234, -110	132, -110	199, -319	272, -110	136, -110	67, -319	290, -214	135, -110	400, -110
b? adv	260, -110	248, -110	212, -110	268, -110	227, -110	132, -319	186, -319	290, -110	153, -110	127, -292	275, -110	153, -110	415, -110
bδ adv	**-64, 0**	-14, 0	24, 0	**-47, 0**	**7, 0**	**0, 209**	13, 0	-18, 0	**-17, 0**	-60, -27	**15, -104**	-18, 0	**-15, 0**
b? ink	283, -110	240, -110	215, -110	276, -110	226, -110	122, -319	199, -319	290, -110	164, -110	49, -292	273, -110	152, -110	421, -110
bδ ink	-87, 0	**-6, 0**	**21, 0**	-55, 0	8, 0	10, 209	**0, 0**	-18, 0	-28, 0	**18, -27**	17, -104	**-17, 0**	-21, 0

Table 4: Lane comparison for a–z 2/2

hex	006E	006F	0070	0071	0072	0073	0074	0075	0076	0077	0078	0079	007A
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
it	*nn̈*	*oȯ*	*pṗ*	*qq̇*	*rṙ*	*sṡ*	*tṫ*	*uu̇*	*vv̇*	*wẘ*	*xẋ*	*yẏ*	*zż*
patch	*nn̈*	*oȯ*	*pṗ*	*qq̇*	*rṙ*	*sṡ*	*tṫ*	*uu̇*	*vv̇*	*wẘ*	*xẋ*	*yẏ*	*zż*
a	394, 646	359, 645	411, 646	369, 645	312, 646	327, 645		386, 646		348, 646	366, 646	367, 644	360, 646
a? adv	396, 645	361, 645	394, 705	383, 645	316, 645	314, 645	318, 775	398, 645	373, 645	481, 645	375, 645	389, 645	355, 645
aδ adv	-2, 1	**-2, 0**	**17, -59**	**-14, 0**	**-4, 1**	13, 0		**-12, 1**		-133, 1	-9, 1	-22, -1	5, 1
a? ink	396, 645	362, 645	382, 705	398, 645	317, 645	321, 645	312, 775	402, 645	363, 645	471, 645	371, 645	378, 645	358, 645
aδ ink	-2, 1	-3, 0	29, -59	-29, 0	-5, 1	**6, 0**		-16, 1		**-123, 1**	**-5, 1**	**-11, -1**	**2, 1**
b	235, -111	186, -107	248, -110	169, -110	113, -110	169, -111	133, -107	226, -111		355, -110	224, -110	368, -107	187, -111
b? adv	236, -110	200, -110	177, -319	178, -319	155, -110	153, -110	130, -110	237, -110	213, -110	321, -110	214, -110	184, -319	195, -110
bδ adv	-1, -1	**-14, 3**	**71, 209**	**-9, 209**	**-42, 0**	16, -1	**3, 3**	**-11, -1**		**34, 0**	**10, 0**	**184, 212**	**-8, -1**
b? ink	236, -110	202, -110	164, -319	193, -319	157, -110	161, -110	124, -110	242, -110	203, -110	311, -110	211, -110	173, -319	198, -110
bδ ink	-1, -1	-16, 3	84, 209	-24, 209	-44, 0	**8, -1**	9, 3	-16, -1		44, 0	13, 0	195, 212	-11, -1
si	***nn̈***	***oȯ***	***pṗ***	***qq̇***	***rṙ***	***sṡ***	***tṫ***	***uu̇***	***vv̇***	***wẘ***	***xẋ***	***yẏ***	***zż***
patch	***nn̈***	***oȯ***	***pṗ***	***qq̇***	***rṙ***	***sṡ***	***tṫ***	***uu̇***	***vv̇***	***wẘ***	***xẋ***	***yẏ***	***zż***
a	264, 646	238, 645	271, 646	238, 645	191, 646	326, 645		255, 645	256, 646	345, 646		262, 645	220, 646
a? adv	284, 645	235, 645	257, 679	267, 645	203, 645	190, 645	189, 765	262, 645	220, 645	332, 645	240, 645	236, 645	210, 645
aδ adv	-20, 1	**3, 0**	**14, -33**	**-29, 0**	-12, 1	136, 0		-7, 0	**36, 1**	**13, 1**		**26, 0**	10, 1
a? ink	283, 645	231, 645	239, 679	269, 645	202, 645	196, 645	184, 765	261, 645	218, 645	326, 645	238, 645	227, 645	213, 645
aδ ink	**-19, 1**	7, 0	32, -33	-31, 0	**-11, 1**	**130, 0**		**-6, 0**	38, 1	19, 1		35, 0	**7, 1**
b	265, -110	243, -110	179, -319	348, -319	135, -110	196, -110	133, -107	239, -110	241, -110	352, -110		242, -319	207, -110
b? adv	293, -110	243, -110	216, -319	233, -319	212, -110	198, -110	173, -110	270, -110	228, -110	341, -110	248, -110	202, -319	218, -110
bδ adv	-28, 0	**0, 0**	-37, 0	115, 0	-77, 0	**-2, 0**	-40, 3	-31, 0	**13, 0**	**11, 0**		**40, 0**	**-11, 0**
b? ink	292, -110	240, -110	198, -319	235, -319	211, -110	205, -110	168, -110	270, -110	227, -110	335, -110	247, -110	193, -319	222, -110
bδ ink	**-27, 0**	3, 0	**-19, 0**	**113, 0**	**-76, 0**	-9, 0	**-35, 3**	-31, 0	14, 0	17, 0		49, 0	-15, 0

Table 5: Lane comparison for Latin and IPA 1/8

hex	00C6	00D0	00D8	00DE	00DF	00E6	00F0	00F8	00FE	0110	0111	0126	0127	0131	0141	0142	014A
cat	Lu	Lu	Lu	Lu	Ll	Ll	Ll	Ll	Ll	Lu	Ll	Lu	Ll	Ll	Lu	Ll	Lu
it	*ÆȦE̱*	*ÐḊḌ*	*ØȮǪ*	*ÞṖ�example*	*ßßß*	*æȧę*	*ððḏ*	*øȯǫ*	*þṗḭ*	*ÐḊḌ*	*đḋḍ*	*ĦḢḤ*	*ħḣḥ*	*ıiᵢ*	*ŁĿḶ*	*łŀḷ*	*ŋṅȵ*
patch	*ÆȦE̱*	*ÐḊḌ*	*ØȮǪ*	*ÞṖ�example*	*ßßß*	*æȧę*	*ððḏ*	*øȯǫ*	*þṗḭ*	*ÐḊḌ*	*đḋḍ*	*ĦḢḤ*	*ħḣḥ*	*ıiᵢ*	*ŁĿḶ*	*łŀḷ*	*ŋṅȵ*
a	615, 850					480, 645			278, 646					267, 645			346, 804
a? adv	595, 850	514, 850	515, 850	431, 850	461, 890	481, 645	429, 896	370, 691	439, 890	514, 850	434, 890	517, 850	449, 890	275, 645	440, 850	322, 890	495, 850
aδ adv	**20, 0**					**-1, 0**			-161, -244					**-8, 0**			-149, -46
a? ink	581, 850	494, 850	510, 850	411, 850	418, 890	485, 645	431, 896	372, 691	429, 890	494, 850	451, 890	510, 850	455, 890	286, 645	422, 850	322, 890	471, 850
aδ ink	34, 0					-5, 0			**-151, -244**					-19, 0			**-125, -46**
b	549, -107		310, -113			327, -109		186, -107						139, -108			364, -109
b? adv	391, -110	310, -110	302, -151	227, -110	204, -319	320, -110	216, -110	190, -156	181, -324	310, -110	221, -110	313, -110	236, -110	115, -110	236, -110	110, -110	253, -285
bδ adv	**158, 3**		**8, 38**			7, 1		**-4, 49**						24, 2			**111, 176**
b? ink	377, -110	290, -110	297, -151	207, -110	161, -319	325, -110	218, -110	192, -156	171, -324	290, -110	239, -110	306, -110	243, -110	126, -110	218, -110	110, -110	229, -285
bδ ink	172, 3		13, 38			**2, 1**		-6, 49						**13, 2**			135, 176
si	***ÆȦE̱***	***ÐḊḌ***	***ØȮǪ***	***ÞṖ�example***	***ßßß***	***æȧę***	***ððḏ***	***øȯǫ***	***þṗḭ***	***ÐḊḌ***	***đḋḍ***	***ĦḢḤ***	***ħḣḥ***	***ıiᵢ***	***ŁĿḶ***	***łŀḷ***	***ŋṅȵ***
patch	***ÆȦE̱***	***ÐḊḌ***	***ØȮǪ***	***ÞṖṖ***	***ßßß***	***æȧę***	***ððḏ***	***øȯǫ***	***þṗḭ***	***ÐḊḌ***	***đḋḍ***	***ĦḢḤ***	***ħḣḥ***	***ıiᵢ***	***ŁĿḶ***	***łŀḷ***	***ŋṅȵ***
a	500, 850	341, 850	351, 850	281, 850		350, 645			278, 646					132, 645			359, 804
a? adv	459, 850	367, 850	390, 850	303, 850	325, 890	339, 645	291, 909	235, 645	300, 890	367, 850	309, 890	396, 850	331, 890	144, 645	300, 850	190, 890	390, 850
aδ adv	**41, 0**	-26, 0	**-39, 0**	-22, 0		11, 0			-22, -244					**-12, 0**			-31, -46
a? ink	448, 850	347, 850	393, 850	285, 850	271, 890	350, 645	298, 909	236, 645	287, 890	333, 850	318, 890	415, 850	331, 890	157, 645	322, 850	207, 890	375, 850
aδ ink	52, 0	**-6, 0**	-42, 0	**-4, 0**		**0, 0**			**-9, -244**					-25, 0			**-16, -46**
b	549, -107		350, -110			327, -109		243, -110						139, -108			373, -109
b? adv	426, -110	334, -110	349, -148	270, -110	241, -319	347, -110	246, -110	237, -138	216, -319	334, -110	268, -110	363, -110	290, -110	153, -110	267, -110	148, -110	325, -268
bδ adv	**123, 3**		**1, 38**			**-20, 1**		6, 28						**-14, 2**			**48, 159**
b? ink	415, -110	314, -110	352, -148	252, -110	187, -319	359, -110	253, -110	239, -138	203, -319	300, -110	277, -110	382, -110	290, -110	166, -110	289, -110	166, -110	309, -268
bδ ink	134, 3		-2, 38			-32, 1		**4, 28**						-27, 2			64, 159

Table 6: Lane comparison for Latin and IPA 2/8

hex	014B	0152	0153	017F	0181	0182	0184	0186	0187	0189	018A	018B	018E	018F	0190	0191	0192
cat	Ll	Lu	Ll	Ll	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Ll
it	*ŋ̇ŋ̇ŋ*	*ŒĖŒ*	*œėœ*	*ẛẛſ*	*ƁḂƁ*	*ƂƂƂ*	*ƄḄƄ*	*ƆƆƆ*	*ƇƇƇ*	*ƉƉƉ*	*ƊḊƊ*	*ƋƋƋ*	*ƎƎƎ*	*ƏƏƏ*	*ƐƐƐ*	*ƑƑƑ*	*ƒƒƒ*
patch	*ŋ̇ŋ̇ŋ*	*ŒĖŒ*	*œėœ*	*ẛẛſ*	*ƁḂƁ*	*ƂƂƂ*	*ƄḄƄ*	*ƆƆƆ*	*ƇƇƇ*	*ƉƉƉ*	*ƊḊƊ*	*ƋƋƋ*	*ƎƎƎ*	*ƏƏƏ*	*ƐƐƐ*	*ƑƑƑ*	*ƒƒƒ*
a	264, 646	575, 850	480, 645		327, 805		267, 805				306, 805						
a? adv	382, 645	587, 850	490, 645	339, 890	488, 850	465, 850	455, 850	503, 850	503, 850	534, 850	553, 850	450, 850	465, 850	515, 850	439, 850	419, 850	414, 890
aδ adv	-118, 1	-12, 0	-10, 0		-161, -45		-188, -45				-247, -45						
a? ink	368, 645	585, 850	489, 645	360, 890	466, 850	454, 850	445, 850	512, 850	560, 850	520, 850	527, 850	459, 850	478, 850	527, 850	431, 850	364, 850	402, 890
aδ ink	**-104, 1**	**-10, 0**	**-9, 0**		**-139, -45**		**-178, -45**				**-221, -45**						
b	265, -111				328, -108		268, -108				369, -107						
b? adv	177, -319	383, -110	329, -110	82, -319	284, -110	261, -110	251, -110	299, -110	299, -110	330, -110	349, -110	246, -110	261, -110	311, -110	235, -110	184, -257	170, -257
bδ adv	**88, 208**				**44, 2**		**17, 2**				**20, 3**						
b? ink	163, -319	381, -110	329, -110	103, -319	262, -110	250, -110	241, -110	308, -110	356, -110	316, -110	323, -110	255, -110	274, -110	323, -110	227, -110	128, -257	158, -257
bδ ink	102, 208				66, 2		27, 2				46, 3						
si	***ŋ̇ŋ̇ŋ***	***ŒĖŒ***	***œėœ***	***ẛẛſ***	***ƁḂƁ***	***ƂƂƂ***	***ƄḄƄ***	*ƆƆƆ*	*ƇƇƇ*	***ƉƉƉ***	***ƊḊƊ***	***ƋƋƋ***	***ƎƎƎ***	***ƏƏƏ***	***ƐƐƐ***	***ƑƑƑ***	***ƒƒƒ***
patch	***ŋ̇ŋ̇ŋ***	***ŒĖŒ***	***œėœ***	***ẛẛſ***	***ƁḂƁ***	***ƂƂƂ***	***ƄḄƄ***	*ƆƆƆ*	*ƇƇƇ*	***ƉƉƉ***	***ƊḊƊ***	***ƋƋƋ***	***ƎƎƎ***	***ƏƏƏ***	***ƐƐƐ***	***ƑƑƑ***	***ƒƒƒ***
a	264, 646	480, 850	415, 645		327, 805		267, 805				306, 805						
a? adv	253, 645	450, 850	384, 645	209, 903	359, 850	303, 850	306, 850	357, 850	359, 850	388, 850	424, 850	304, 850	320, 850	369, 850	293, 850	288, 850	269, 890
aδ adv	**11, 1**	30, 0	31, 0		-32, -45		-39, -45				-118, -45						
a? ink	236, 645	464, 850	392, 645	222, 903	329, 850	279, 850	289, 850	337, 850	398, 850	382, 850	393, 850	313, 850	333, 850	368, 850	287, 850	254, 850	259, 890
aδ ink	28, 1	**16, 0**	**23, 0**		**-2, -45**		**-22, -45**				**-87, -45**						
b	265, -111				328, -108		268, -108				369, -107						
b? adv	222, -301	417, -110	393, -110	121, -326	326, -110	270, -110	273, -110	324, -110	326, -110	355, -110	391, -110	271, -110	287, -110	336, -110	260, -110	220, -281	198, -255
bδ adv	**43, 190**				**2, 2**		**-5, 2**				-22, 3						
b? ink	206, -301	431, -110	401, -110	134, -326	296, -110	246, -110	256, -110	304, -110	365, -110	349, -110	360, -110	280, -110	300, -110	335, -110	254, -110	186, -281	188, -255
bδ ink	59, 190				32, 2		12, 2				**9, 3**						

Table 7: Lane comparison for Latin and IPA 3/8

hex	0193	0194	0196	0197	0198	019C	019D	019F	01A2	01A9	01B7	0237	024A	0250	0251	0252	0253
cat	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Lu	Ll	Lu	Ll	Ll	Ll	Ll
it	*Ɠ̈Ɠ̱*	*Ɣ̈Ɣ̱*	*Ɩ̈Ɩ̱*	*Ɨ̈Ɨ̱*	*Ƙ̈Ƙ̱*	*Ɯ̈Ɯ̱*	*Ɲ̈Ɲ̱*	*Ɵ̈Ɵ̱*	*Ơ̈Ơ̱*	*Ʃ̈Ʃ̱*	*Ʒ̈Ʒ̱*	*ȷ̈ȷ̱*	▨	*ɐ̈ɐ̱*	*ɑ̈ɑ̱*	*ɒ̈ɒ̱*	*ɓ̈ɓ̱*
patch	*Ɠ̈Ɠ̱*	*Ɣ̈Ɣ̱*	*Ɩ̈Ɩ̱*	*Ɨ̈Ɨ̱*	*Ƙ̈Ƙ̱*	*Ɯ̈Ɯ̱*	*Ɲ̈Ɲ̱*	*Ɵ̈Ɵ̱*	*Ơ̈Ơ̱*	*Ʃ̈Ʃ̱*	*Ʒ̈Ʒ̱*	*ȷ̈ȷ̱*	▨	*ɐ̈ɐ̱*	*ɑ̈ɑ̱*	*ɒ̈ɒ̱*	*ɓ̈ɓ̱*
a		331, 805				431, 801			355, 804			140, 645					
a? adv	541, 850	462, 850	322, 890	321, 850	499, 850	609, 850	528, 850	515, 850	622, 850	477, 850	485, 850	267, 645		365, 645	380, 645	391, 645	431, 896
aδ adv		-131, -45				-178, -49			**-267, -46**			-127, 0					
a? ink	537, 850	439, 850	327, 890	315, 850	494, 850	600, 850	496, 850	519, 850	632, 850	444, 850	485, 850	205, 645		361, 645	404, 645	379, 645	448, 896
aδ ink		**-108, -45**				**-169, -49**			-277, -46			**-65, 0**					
b						450, -108			350, -113					229, -114	212, -114	243, -114	212, -114
b? adv	337, -110	218, -297	110, -110	117, -110	295, -110	405, -110	290, -272	311, -110	393, -228	273, -110	241, -296	70, -281		205, -110	220, -110	231, -110	218, -110
bδ adv						**45, 2**			**-43, 115**					**24, -4**	**-8, -4**	**12, -4**	**-6, -4**
b? ink	333, -110	195, -297	115, -110	111, -110	290, -110	396, -110	257, -272	315, -110	403, -228	240, -110	241, -296	8, -281		201, -110	244, -110	219, -110	235, -110
bδ ink						54, 2			-53, 115					28, -4	-32, -4	24, -4	-23, -4
si	***Ɠ̈Ɠ̱***	***Ɣ̈Ɣ̱***	***Ɩ̈Ɩ̱***	***Ɨ̈Ɨ̱***	***Ƙ̈Ƙ̱***	***Ɯ̈Ɯ̱***	***Ɲ̈Ɲ̱***	***Ɵ̈Ɵ̱***	***Ơ̈Ơ̱***	***Ʃ̈Ʃ̱***	***Ʒ̈Ʒ̱***	***ȷ̈ȷ̱***	▨	***ɐ̈ɐ̱***	***ɑ̈ɑ̱***	***ɒ̈ɒ̱***	***ɓ̈ɓ̱***
patch	***Ɠ̈Ɠ̱***	***Ɣ̈Ɣ̱***	***Ɩ̈Ɩ̱***	***Ɨ̈Ɨ̱***	***Ƙ̈Ƙ̱***	***Ɯ̈Ɯ̱***	***Ɲ̈Ɲ̱***	***Ɵ̈Ɵ̱***	***Ơ̈Ơ̱***	***Ʃ̈Ʃ̱***	***Ʒ̈Ʒ̱***	***ȷ̈ȷ̱***	▨	***ɐ̈ɐ̱***	***ɑ̈ɑ̱***	***ɒ̈ɒ̱***	***ɓ̈ɓ̱***
a		331, 805				431, 801			359, 804			140, 645			228, 645	264, 645	
a? adv	395, 850	316, 850	194, 890	181, 850	353, 850	463, 850	383, 850	387, 850	464, 850	319, 850	329, 850	121, 645		221, 645	247, 645	247, 645	284, 890
aδ adv		**15, -45**				-32, -49			**-105, -46**			**19, 0**			**-19, 0**	**17, 0**	
a? ink	396, 850	299, 850	193, 890	185, 850	352, 850	458, 850	353, 850	394, 850	474, 850	290, 850	326, 850	89, 645		203, 645	262, 645	239, 645	302, 890
aδ ink		32, -45				**-27, -49**			-115, -46			51, 0			-34, 0	25, 0	
b						450, -108			354, -113		236, -319			229, -114	212, -114	259, -114	212, -114
b? adv	362, -110	245, -295	153, -110	148, -110	320, -110	430, -110	317, -270	354, -110	407, -227	286, -110	258, -293	93, -292		230, -110	256, -110	256, -110	243, -110
bδ adv						**20, 2**			**-53, 114**		-22, -26			**-1, -4**	**-44, -4**	**3, -4**	**-31, -4**
b? ink	363, -110	228, -295	152, -110	152, -110	319, -110	425, -110	287, -270	361, -110	417, -227	257, -110	255, -293	61, -292		212, -110	271, -110	248, -110	261, -110
bδ ink						25, 2			-63, 114		**-19, -26**			17, -4	-59, -4	11, -4	-49, -4

Table 8: Lane comparison for Latin and IPA 4/8

hex	0254	0255	0256	0257	0258	0259	025A	025B	025C	025D	025E	025F	0260	0261	0262	0263	0264
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
it	ɔɔ̣	ɕɕ̣	ɖɖ̣	ɗɗ̣	ɞɞ̣	əə̣	ɚɚ̣	ɛɛ̣	ɜɜ̣	ɝɝ̣	ɵɵ̣	ɟɟ̣	ɠɠ̣	ɡɡ̣	ɢɢ̣	ɣɣ̣	ɤɤ̣
patch	ɔɔ̣	ɕɕ̣	ɖɖ̣	ɗɗ̣	ɞɞ̣	əə̣	ɚɚ̣	ɛɛ̣	ɜɜ̣	ɝɝ̣	ɵɵ̣	ɟɟ̣	ɠɠ̣	ɡɡ̣	ɢɢ̣	ɣɣ̣	ɤɤ̣
a	322, 654		218, 645	218, 645	204, 645			322, 654							265, 645		
a? adv	332, 645	352, 645	465, 899	464, 890	356, 645	356, 645	418, 687	338, 645	331, 645	413, 775	375, 645	305, 645	431, 896	378, 645	388, 645	387, 645	338, 645
aδ adv	-10, 9		**-247, -254**	**-246, -245**	**-152, 0**			**-16, 9**							-123, 0		
a? ink	327, 645	355, 645	500, 899	502, 890	359, 645	358, 645	459, 687	340, 645	331, 645	425, 775	376, 645	273, 645	521, 896	366, 645	386, 645	393, 645	345, 645
aδ ink	**-5, 9**		-282, -254	-284, -245	-155, 0			-18, 9							**-121, 0**		
b	172, -114	212, -114	212, -114	212, -114	212, -114	212, -114	212, -114	212, -114	184, -114	212, -114	256, -114				248, -114		177, -114
b? adv	171, -110	171, -208	215, -273	251, -110	195, -110	195, -110	248, -110	178, -110	170, -110	225, -110	209, -133	100, -319	180, -286	180, -286	227, -110	192, -272	178, -106
bδ adv	**1, -4**	41, 94	**-3, 159**	**-39, -4**	17, -4	17, -4	**-36, -4**	34, -4	14, -4	**-13, -4**	47, 19				**21, -4**		**-1, -8**
b? ink	167, -110	174, -208	251, -273	290, -110	199, -110	198, -110	290, -110	180, -110	171, -110	237, -110	211, -133	68, -319	270, -286	168, -286	226, -110	198, -272	185, -106
bδ ink	5, -4	**38, 94**	-39, 159	-78, -4	**13, -4**	**14, -4**	-78, -4	**32, -4**	**13, -4**	-25, -4	**45, 19**				22, -4		-8, -8
si	***ɔɔ̣***	***ɕɕ̣***	***ɖɖ̣***	***ɗɗ̣***	***ɞɞ̣***	***əə̣***	***ɚɚ̣***	***ɛɛ̣***	***ɜɜ̣***	***ɝɝ̣***	***ɵɵ̣***	***ɟɟ̣***	***ɠɠ̣***	***ɡɡ̣***	***ɢɢ̣***	***ɣɣ̣***	***ɤɤ̣***
patch	***ɔɔ̣***	***ɕɕ̣***	***ɖɖ̣***	***ɗɗ̣***	***ɞɞ̣***	***əə̣***	***ɚɚ̣***	***ɛɛ̣***	***ɜɜ̣***	***ɝɝ̣***	***ɵɵ̣***	***ɟɟ̣***	***ɠɠ̣***	***ɡɡ̣***	***ɢɢ̣***	***ɣɣ̣***	***ɤɤ̣***
a	160, 645		218, 645	218, 645	204, 645			209, 645	194, 645						265, 645		
a? adv	199, 645	204, 645	302, 890	318, 890	207, 645	218, 645	299, 693	194, 645	194, 645	261, 762	232, 645	141, 645	309, 890	238, 645	244, 645	255, 645	185, 645
aδ adv	-39, 0		**-84, -245**	**-100, -245**	**-3, 0**			15, 0	**0, 0**						21, 0		
a? ink	184, 645	220, 645	355, 890	366, 890	215, 645	212, 645	319, 693	197, 645	199, 645	288, 762	237, 645	126, 645	378, 890	225, 645	252, 645	264, 645	195, 645
aδ ink	**-24, 0**		-137, -245	-148, -245	-11, 0			**12, 0**	-5, 0						**13, 0**		
b	177, -114	212, -114	212, -114	212, -114	212, -114		212, -114	212, -114	184, -114	212, -114	256, -114				248, -110		177, -114
b? adv	207, -110	192, -209	227, -271	276, -110	216, -110	227, -110	298, -110	203, -110	203, -110	246, -110	241, -110	107, -319	232, -284	204, -319	252, -110	228, -285	194, -110
bδ adv	-30, -4	20, 95	**-15, 157**	**-64, -4**	**-4, -4**		**-86, -4**	9, -4	**-19, -4**	**-34, -4**	15, -4				**-4, 0**		**-17, -4**
b? ink	193, -110	208, -209	281, -271	325, -110	224, -110	221, -110	318, -110	206, -110	208, -110	273, -110	246, -110	92, -319	301, -284	191, -319	261, -110	237, -285	204, -110
bδ ink	**-16, -4**	**4, 95**	-69, 157	-113, -4	-12, -4		-106, -4	**6, -4**	-24, -4	-61, -4	**10, -4**				-13, 0		-27, -4

Table 9: Lane comparison for Latin and IPA 5/8

hex	0265	0266	0267	0268	0269	026A	026B	026C	026D	026E	026F	0270	0271	0272	0273	0274	0275
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
it	*ɥ̡ɥ̡ɥ*	*ɦ̡ɦ̡ɦ*	*ɧ̡ɧ̡ɧ*	*ɨ̡ɨ̡ɨ*	*ᵢ̡ᵢ*	*ɪ̡ɪ*	*ɬ̡ɬ*	*ꞎ̡ꞎ*	*ɭ̡ɭ*	*ɮ̡ɮ*	*ɯ̡ɯ*	*ɰ̡ɰ*	*ŋ̡ŋ*	*ɲ̡ɲ*	*ɳ̡ɳ*	*ɴ̡ɴ*	*ɵ̡ɵ*
patch	*ɥ̡ɥ̲*	*ɦ̡ɦ̲*	*ɧ̡ɧ̲*	*ɨ̡ɨ̡ɨ*	*ᵢ̡ᵢ*	*ɪ̡ɪ*	*ɬ̡ɬ*	*ꞎ̡ꞎ*	*ɭ̡ɭ*	*ɮ̡ɮ*	*ɯ̡ɯ*	*ɰ̡ɰ*	*ŋ̡ŋ*	*ɲ̡ɲ*	*ɳ̡ɳ*	*ɴ̡ɴ*	*ɵ̡ɵ*
a	258, 645	299, 565			123, 645	128, 645					407, 645		383, 645	264, 646	263, 645		
a? adv	396, 645	449, 890	449, 890	316, 810	266, 645	281, 645	333, 890	368, 890	341, 890	447, 890	517, 645	515, 645	515, 645	396, 645	367, 645	418, 645	379, 645
aδ adv	**-138, 0**	**-150, -325**			**-143, 0**	-153, 0					**-110, 0**		-132, 0	-132, 1	**-104, 0**		
a? ink	402, 645	487, 890	460, 890	324, 810	297, 645	277, 645	331, 890	367, 890	354, 890	445, 890	521, 645	526, 645	489, 645	362, 645	416, 645	420, 645	381, 645
aδ ink	-144, 0	-188, -325			-174, 0	**-149, 0**					-114, 0		**-106, 0**	**-98, 1**	-153, 0		
b	219, -129	240, -114		134, -114	153, -114	139, -114	146, -114	218, -120			410, -114	294, -114	383, -105	265, -111	262, -112	247, -114	
b? adv	191, -319	236, -110	192, -319	121, -110	106, -110	120, -110	120, -110	156, -110	91, -287	192, -309	354, -123	310, -319	310, -319	191, -319	162, -319	258, -110	218, -110
bδ adv	28, 190	**4, -4**		13, -4	47, -4	**19, -4**	**26, -4**	**62, -10**			56, 9	**-16, 205**	**73, 214**	**74, 208**	100, 207	**-11, -4**	
b? ink	197, -319	275, -110	203, -319	129, -110	137, -110	117, -110	119, -110	155, -110	104, -287	190, -309	358, -123	321, -319	284, -319	157, -319	211, -319	260, -110	221, -110
bδ ink	**22, 190**	-35, -4		**5, -4**	**16, -4**	22, -4	27, -4	63, -10			**52, 9**	-27, 205	99, 214	108, 208	**51, 207**	-13, -4	
si	***ɥ̡ɥ̡ɥ***	***ɦ̡ɦ̡ɦ***	***ɧ̡ɧ̡ɧ***	***ɨ̡ɨ̡ɨ***	***ᵢ̡ᵢ***	***ɪ̡ɪ***	***ɬ̡ɬ***	***ꞎ̡ꞎ***	***ɭ̡ɭ***	***ɮ̡ɮ***	***ɯ̡ɯ***	***ɰ̡ɰ***	***ŋ̡ŋ***	***ɲ̡ɲ***	***ɳ̡ɳ***	***ɴ̡ɴ***	***ɵ̡ɵ***
patch	***ɥ̡ɥ̲***	***ɦ̡ɦ̲***	***ɧ̡ɧ̲***	***ɨ̡ɨ̡ɨ***	***ᵢ̡ᵢ***	***ɪ̡ɪ***	***ɬ̡ɬ***	***ꞎ̡ꞎ***	***ɭ̡ɭ***	***ɮ̡ɮ***	***ɯ̡ɯ***	***ɰ̡ɰ***	***ŋ̡ŋ***	***ɲ̡ɲ***	***ɳ̡ɳ***	***ɴ̡ɴ***	***ɵ̡ɵ***
a	258, 645	299, 565	299, 565	137, 795	123, 645	128, 645					407, 645		383, 645	264, 646	263, 645		
a? adv	258, 645	302, 890	301, 890	180, 823	138, 645	137, 645	187, 890	223, 890	187, 890	301, 890	372, 640	373, 645	373, 645	254, 645	254, 645	275, 645	229, 645
aδ adv	**0, 0**	**-3, -325**	**-2, -325**	**-43, -28**	**-15, 0**	-9, 0					**35, 5**		**10, 0**	**10, 1**	**9, 0**		
a? ink	267, 645	305, 890	278, 890	191, 823	152, 645	136, 645	220, 890	234, 890	243, 890	299, 890	368, 640	381, 645	348, 645	224, 645	285, 645	277, 645	258, 645
aδ ink	-9, 0	-6, -325	21, -325	-54, -28	-29, 0	**-8, 0**					39, 5		35, 0	40, 1	-22, 0		
b	219, -129	240, -114	240, -114	136, -110	153, -114	139, -114	146, -114	218, -120			410, -114	294, -114	383, -105	265, -111	262, -112	247, -114	
b? adv	224, -319	260, -110	217, -319	153, -110	147, -110	145, -110	145, -110	181, -110	103, -319	217, -319	382, -110	339, -319	339, -319	220, -319	220, -319	283, -110	238, -110
bδ adv	**-5, 190**	**-20, -4**	**23, 205**	**-17, 0**	**6, -4**	-6, -4	**1, -4**	37, -10			**28, -4**	**-45, 205**	**44, 214**	**45, 208**	42, 207	**-36, -4**	
b? ink	233, -319	264, -110	194, -319	164, -110	161, -110	145, -110	179, -110	193, -110	159, -319	215, -319	378, -110	347, -319	314, -319	190, -319	251, -319	286, -110	267, -110
bδ ink	-14, 190	-24, -4	46, 205	-28, 0	-8, -4	-6, -4	-33, -4	**25, -10**			32, -4	-53, 205	69, 214	75, 208	**11, 207**	-39, -4	

Table 10: Lane comparison for Latin and IPA 6/8

hex	0276	0277	0278	0279	027A	027B	027C	027D	027E	027F	0280	0281	0282	0283	0284	0285	0286
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll
it	ɐɛɘɜ	ɷɷɷ	ɸɸɸ	ɾɾɾ	ɽɽɽ	ɾɾɾ	ɻɻɻ	ɻɻɻ	ɾɾɾ	ɾɾɾ	ʁʁʁ	ʁʁʁ	ʂʂʂ	ʃʃʃ	ʃʃʃ	ɭɭɭ	ʃʃʃ
patch	ɐɛɘɜ	ɷɷɷ	ɸɸɸ	ɾɾɾ	ɽɽɽ	ɾɾɾ	ɻɻɻ	ɻɻɻ	ɾɾɾ	ɾɾɾ	ʁʁʁ	ʁʁʁ	ʂʂʂ	ʃʃʃ	ʃʃʃ	ɳɳɳ	ʃʃʃ
a							222, 703										
a? adv	457, 645	494, 645	481, 890	315, 645	362, 819	352, 645	325, 645	320, 645	311, 645	307, 645	358, 645	349, 645	314, 645	343, 900	343, 900	343, 900	343, 900
aδ adv							**-103, 58**										
a? ink	462, 645	497, 645	485, 890	223, 645	373, 819	362, 645	332, 645	331, 645	315, 645	311, 645	361, 645	351, 645	328, 645	336, 900	336, 900	335, 900	345, 900
aδ ink							-110, 58										
b		341, -114		190, -114	203, -114		127, -305										
b? adv	296, -110	330, -126	231, -285	155, -110	161, -127	160, -259	128, -283	124, -279	151, -110	147, -110	197, -110	189, -110	121, -265	88, -299	88, -299	88, -299	92, -283
bδ adv		11, 12		**35, -4**	42, 13		**-1, -22**										
b? ink	302, -110	333, -126	235, -285	63, -110	172, -127	170, -259	135, -283	135, -279	155, -110	151, -110	201, -110	191, -110	135, -265	81, -299	81, -299	80, -299	94, -283
bδ ink		**8, 12**		127, -4	**31, 13**		-8, -22										
si	**ɐɛɘɜ**	**ɷɷɷ**	**ɸɸɸ**	**ɾɾɾ**	**ɽɽɽ**	**ɾɾɾ**	**ɻɻɻ**	**ɻɻɻ**	**ɾɾɾ**	**ɾɾɾ**	**ʁʁʁ**	**ʁʁʁ**	**ʂʂʂ**	**ʃʃʃ**	**ʃʃʃ**	**ɭɭɭ**	**ʃʃʃ**
patch	**ɐɛɘɜ**	**ɷɷɷ**	**ɸɸɸ**	**ɾɾɾ**	**ɽɽɽ**	**ɾɾɾ**	**ɻɻɻ**	**ɻɻɻ**	**ɾɾɾ**	**ɾɾɾ**	**ʁʁʁ**	**ʁʁʁ**	**ʂʂʂ**	**ʃʃʃ**	**ʃʃʃ**	**ɳɳɳ**	**ʃʃʃ**
a							222, 703										
a? adv	313, 642	350, 645	328, 890	171, 640	217, 821	194, 645	181, 645	177, 645	167, 645	167, 645	214, 642	214, 645	178, 645	195, 890	195, 890	195, 890	189, 890
aδ adv							41, 58										
a? ink	320, 642	356, 645	334, 890	158, 640	231, 821	218, 645	190, 645	190, 645	173, 645	169, 645	217, 642	210, 645	202, 645	190, 890	190, 890	190, 890	199, 890
aδ ink							**32, 58**										
b		341, -114		190, -114	203, -114		127, -305										
b? adv	322, -110	359, -110	251, -283	180, -110	190, -110	172, -257	155, -281	143, -319	176, -110	176, -110	223, -110	223, -110	155, -265	116, -296	116, -296	116, -296	105, -319
bδ adv		**-18, -4**		**10, -4**	13, -4		**-28, -24**										
b? ink	329, -110	365, -110	257, -283	168, -110	204, -110	197, -257	164, -281	156, -319	182, -110	178, -110	226, -110	219, -110	179, -265	111, -296	111, -296	111, -296	115, -319
bδ ink		-24, -4		22, -4	**-1, -4**		-37, -24										

Table 11: Lane comparison for Latin and IPA 7/8

hex	0287	0288	0289	028A	028B	028C	028D	028E	028F	0290	0291	0292	0293	0294	0295	0296	0297
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Lu	Lu	Lu	Ll
it	ɪ̡ɪ̡	t̡t̡	ʉ̡ʉ̡	ʊ̡ʊ̡	ʋ̡ʋ̡	ʌ̡ʌ̡	ɯ̡ɯ̡	ʏ̡ʏ̡	ɤ̡ɤ̡	ʐ̡ʐ̡	ʑ̡ʑ̡	ʒ̡ʒ̡	ʒ̡ʒ̡	ʔ̡ʔ̡	ʕ̡ʕ̡	ɧ̡ɧ̡	ɕ̡ɕ̡
patch	ɪ̡ɪ̡	t̡t̡	ʉ̡ʉ̡	ʊ̡ʊ̡	ʋ̡ʋ̡	ʌ̡ʌ̡	ɯ̡ɯ̡	ʏ̡ʏ̡	ɤ̡ɤ̡	ʐ̡ʐ̡	ʑ̡ʑ̡	ʒ̡ʒ̡	ʒ̡ʒ̡	ʔ̡ʔ̡	ʕ̡ʕ̡	ɧ̡ɧ̡	ɕ̡ɕ̡
a					259, 645							328, 646					
a? adv	288, 645	314, 776	398, 645	392, 645	382, 645	383, 645	506, 645	433, 838	372, 645	343, 645	352, 645	350, 645	355, 645	373, 850	373, 850	373, 850	411, 645
aδ adv					**-123, 0**							-22, 1					
a? ink	300, 645	337, 776	408, 645	396, 645	385, 645	383, 645	507, 645	430, 838	379, 645	391, 645	356, 645	347, 645	362, 645	384, 850	383, 850	376, 850	402, 645
aδ ink					-126, 0							**-19, 1**					
b				258, -114	251, -114				239, -107			130, -319		202, -107	230, -107		
b? adv	100, -239	83, -311	238, -110	229, -121	218, -124	222, -110	346, -110	231, -110	212, -110	147, -276	171, -207	145, -319	133, -397	169, -110	169, -110	164, -131	206, -319
bδ adv				29, 7	33, 10				27, 3			-15, 0		33, 3	61, 3		
b? ink	112, -239	106, -311	248, -110	233, -121	222, -124	223, -110	347, -110	229, -110	219, -110	195, -276	175, -207	142, -319	141, -397	180, -110	179, -110	167, -131	197, -319
bδ ink				**25, 7**	**29, 10**				**20, 3**			**-12, 0**		**22, 3**	**51, 3**		
si	ɪ̡ɪ̡	t̡t̡	ʉ̡ʉ̡	ʊ̡ʊ̡	ʋ̡ʋ̡	ʌ̡ʌ̡	ɯ̡ɯ̡	ʏ̡ʏ̡	ɤ̡ɤ̡	ʐ̡ʐ̡	ʑ̡ʑ̡	ʒ̡ʒ̡	ʒ̡ʒ̡	ʔ̡ʔ̡	ʕ̡ʕ̡	ɧ̡ɧ̡	ɕ̡ɕ̡
patch	ɪ̡ɪ̡	t̡t̡	ʉ̡ʉ̡	ʊ̡ʊ̡	ʋ̡ʋ̡	ʌ̡ʌ̡	ɯ̡ɯ̡	ʏ̡ʏ̡	ɤ̡ɤ̡	ʐ̡ʐ̡	ʑ̡ʑ̡	ʒ̡ʒ̡	ʒ̡ʒ̡	ʔ̡ʔ̡	ʕ̡ʕ̡	ɧ̡ɧ̡	ɕ̡ɕ̡
a					259, 645							218, 646					
a? adv	149, 645	172, 787	255, 645	248, 642	241, 645	239, 645	363, 645	286, 834	228, 645	202, 645	208, 645	206, 645	211, 645	227, 850	227, 850	227, 850	268, 645
aδ adv					18, 0							**12, 1**					
a? ink	136, 645	195, 787	285, 645	255, 642	246, 645	220, 645	346, 645	273, 834	237, 645	252, 645	213, 645	205, 645	217, 645	244, 850	243, 850	232, 850	258, 645
aδ ink					**13, 0**							13, 1					
b				258, -114	251, -114				239, -107			141, -319		202, -107	230, -107		
b? adv	129, -250	109, -319	264, -110	257, -110	249, -110	247, -110	371, -110	256, -110	237, -110	176, -279	196, -210	172, -319	157, -413	194, -110	194, -110	194, -110	237, -302
bδ adv				**1, -4**	**2, -4**				**2, 3**			-31, 0		**8, 3**	36, 3		
b? ink	116, -250	132, -319	294, -110	264, -110	255, -110	229, -110	355, -110	243, -110	246, -110	226, -279	201, -210	171, -319	164, -413	211, -110	210, -110	199, -110	228, -302
bδ ink				-6, -4	-4, -4				-7, 3			**-30, 0**		-9, 3	**20, 3**		

Table 12: Lane comparison for Latin and IPA 8/8

hex	0299	029B	029C	029D	029F	02AE	02AF	03B2	03B8	03C7	1EFA	2132	2C69	2C6B
cat	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Ll	Lu	Lu	Lu	Lu
it	ʙ̈ʙ̲	ɢ̈ɢ̲	ʜ̈ʜ̲	j̈j̲	ʟ̈ʟ̲	ɥ̈ɥ̲	ɥ̈ɥ̲	β̈β̲	θ̈θ̲	χ̈χ̲	𝒳̈𝒳̲	𝒳̈𝒳̲	K̈K̲	Z̈Z̲
patch	ʙ̈ʙ̲	ɢ̈ɢ̲	ʜ̈ʜ̲	j̈j̲	ʟ̈ʟ̲	ɥ̈ɥ̲	ɥ̈ɥ̲	β̈β̲	θ̈θ̲	χ̈χ̲	𝒳̈𝒳̲	𝒳̈𝒳̲	K̈K̲	Z̈Z̲
a						278, -214	278, -214						336, 798	320, 803
a? adv	352, 645	410, 796	416, 645	303, 810	355, 645	422, 645	433, 645	425, 890	413, 890	372, 645			516, 850	490, 850
aδ adv						**-144, -859**	**-155, -859**						**-180, -52**	**-170, -47**
a? ink	354, 645	450, 796	422, 645	292, 810	345, 645	432, 645	467, 645	428, 890	419, 890	375, 645			521, 850	505, 850
aδ ink						-154, -859	-189, -859						-185, -52	-185, -47
b			273, -110			318, 542	318, 542	275, -114					342, -109	287, -107
b? adv	192, -110	217, -110	256, -110	77, -254	194, -110	220, -305	227, -326	168, -319	200, -110	167, -319			288, -223	265, -207
bδ adv			17, 0			98, 847	91, 868	107, 205					54, 114	22, 100
b? ink	194, -110	258, -110	262, -110	66, -254	185, -110	230, -305	261, -326	171, -319	207, -110	170, -319			293, -223	280, -207
bδ ink			**11, 0**			**88, 847**	**57, 868**	**104, 205**					**49, 114**	**7, 100**
si	***ʙ̈ʙ̲***	***ɢ̈ɢ̲***	***ʜ̈ʜ̲***	***j̈j̲***	***ʟ̈ʟ̲***	***ɥ̈ɥ̲***	***ɥ̈ɥ̲***	***β̈β̲***	***θ̈θ̲***	***χ̈χ̲***	𝒳̈𝒳̲	𝒳̈𝒳̲	***K̈K̲***	***Z̈Z̲***
patch	***ʙ̈ʙ̲***	***ɢ̈ɢ̲***	***ʜ̈ʜ̲***	***j̈j̲***	***ʟ̈ʟ̲***	***ɥ̈ɥ̲***	***ɥ̈ɥ̲***	***β̈β̲***	***θ̈θ̲***	***χ̈χ̲***	𝒳̈𝒳̲	𝒳̈𝒳̲	***K̈K̲***	***Z̈Z̲***
a						278, -214	278, -214						336, 798	320, 803
a? adv	213, 645	278, 796	342, 671	158, 812	211, 645	279, 645	289, 645	302, 890	277, 873	256, 645			358, 850	332, 850
aδ adv						**-1, -859**	**-11, -859**						**-22, -52**	**-12, -47**
a? ink	220, 645	325, 796	344, 671	147, 812	202, 645	292, 645	324, 645	321, 890	286, 873	272, 645			361, 850	345, 850
aδ ink						-14, -859	-46, -859						-25, -52	-25, -47
b			327, -110			318, 542	318, 542	275, -114					342, -109	287, -107
b? adv	221, -110	256, -110	345, -110	103, -252	219, -110	245, -319	255, -319	218, -319	239, -110	231, -275			302, -221	279, -206
bδ adv			**-18, 0**			73, 861	63, 861	57, 205					40, 112	8, 99
b? ink	229, -110	303, -110	347, -110	93, -252	211, -110	258, -319	290, -319	237, -319	248, -110	247, -275			305, -221	292, -206
bδ ink			-20, 0			**60, 861**	**28, 861**	**38, 205**					**37, 112**	**-5, 99**